
Lecture 9: Emission Mechanisms of Continuous Radiation

Chin-Fei Lee

`cflee@asiaa.sinica.edu.tw`

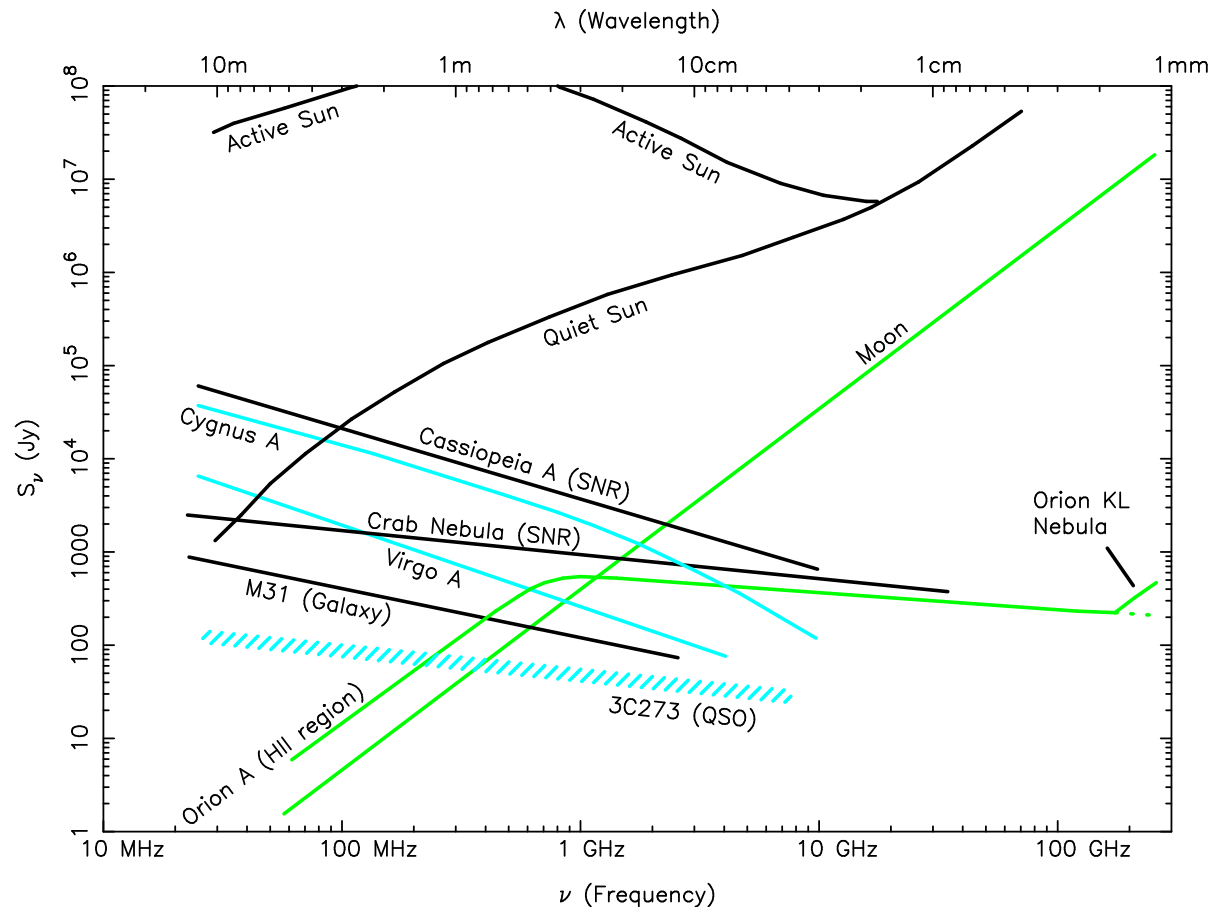
Nature of Radio Sources



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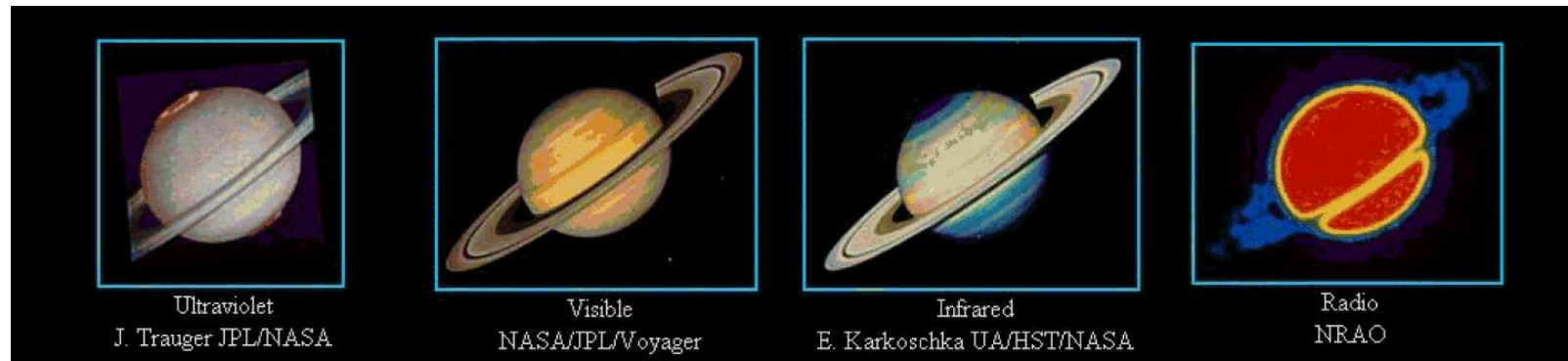
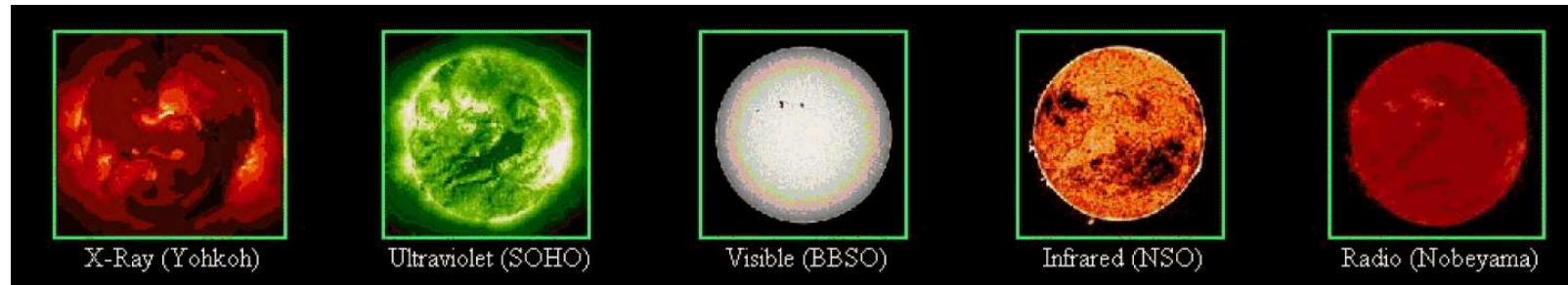
It may appear similar to the starry sky, but in fact most of the point-like objects are not stars, but luminous radio galaxies billions of light years away. The larger sources are ionized clouds of hydrogen, or supernova remnants.

Nature of Radio Sources



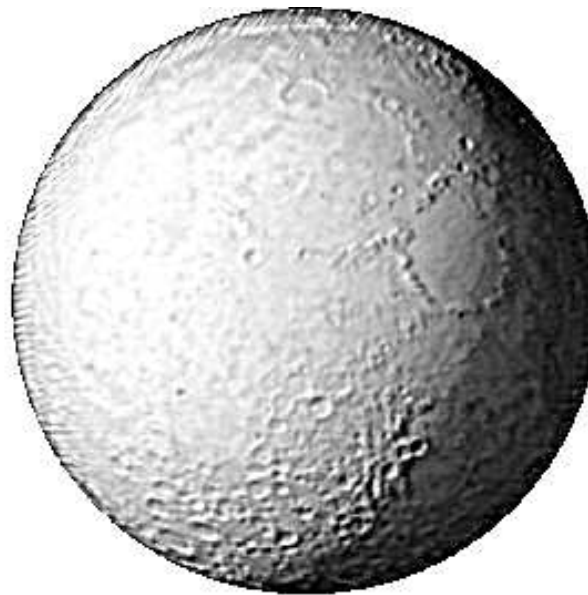
The spectral energy distributions of various radio sources. The Moon, the quiet Sun and (at lower ν) the HII region Orion A (optically thin above 1 GHz) are examples of **Black Bodies**. Close to 300 GHz there is **additional emission from dust** in the molecular cloud Orion KL Nebula. The active Sun, SNRs such as Cassiopeia A, the radio galaxies Cygnus A, Virgo A and the Quasi Stellar radio source (Quasar or QSO) 3C273 are **nonthermal emission**. The hatching indicates rapid time variability. Quasars and radio galaxies are likely the same kind of objects, namely both with AGNs at the centers.

Inside our Solar System



Thermal emission: Blackbody

The first examples of black bodies found were [solar system objects such as the Moon, planets and other solid bodies](#). At radio wavelengths, there is very little reflected sunlight so the radio emission observed from these objects is dominated by thermal emission from their surfaces. Their surfaces are heated by the Sun and radiate in [infrared and radio](#). If the Sun were suddenly switched off, the planets would still remain radio sources for a long time, slowly fading as they cooled.



[Thermal emission from the Moon at \$\lambda = 850 \mu\text{m}\$](#) . The darker right edge ($\sim 100 \text{ K}$) of the Moon is not being illuminated by the Sun, but it still emits radio waves because it does not cool to absolute zero during the lunar night. The brighter side is hotter (200–300 K) and emits more emission.

Thermal emission: Blackbody

For such objects, $\tau = \infty$, so

$$I_\nu = B_\nu(T_s) \equiv \frac{2h\nu^3}{c^2} \frac{1}{e^{h\nu/kT} - 1}$$

and the radio telescope measures the temperature at a depth corresponding to a λ or so. This temperature, referred to as **surface temperature T_s** , is nearly constant with frequency. In the Rayleigh-Jeans limit, $B_\nu^{RJ} \approx \frac{2\nu^2}{c^2} kT_s$ and thus

$$I_\nu = \frac{2\nu^2}{c^2} kT_s \quad (\propto \nu^2 \text{ as seen in the spectrum})$$

and the brightness temperature is

$$T_b \equiv \frac{c^2}{2k\nu^2} I_\nu = T_s$$

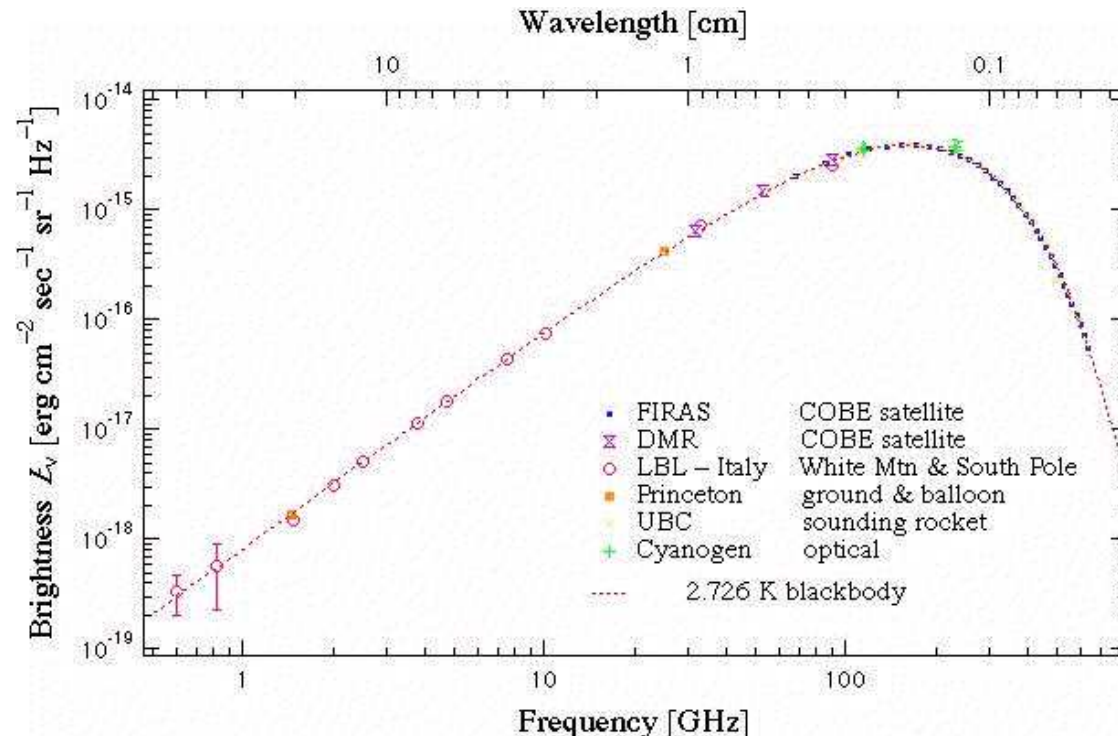
The measured main beam temperature is then

$$T_{\text{MB}} = fT_b = fT_s \quad \text{where} \quad f \equiv \frac{\theta_s^2}{\theta_s^2 + \theta_b^2}$$

where f is the beam filling factor. Once θ_s and thus f are known, T_s can be derived accurately.

Cosmic microwave background

The most famous black body radiation is the 2.7 K microwave background radiation, discovered in 1965. According to the Big Bang scenario, the cosmic microwave background radiation is a relic of the earliest phase of the Universe. The difficulty in detecting it was not due to its weakness, but rather due to the fact that the radiation is present in all directions, so that scanning a telescope over the sky will not lead to a detection. The actual discovery was due to exact measurements of the noise temperature from the sky, the receiver and the ground, compared to the temperature of a helium cooled load. With more advanced instruments, the spectral energy distribution (SED) has been measured beautifully.



Thermal Radiation of Dust

The InterStellar Medium (ISM) is mainly composed of matter in two physical states: [gas and dust](#). Dust absorbs the energy of nearby stars in the ultra violet and optical domains and radiates mainly in the infrared and submillimetre domain. Although it represents only 1% of the gas mass, it is responsible for around 30% of the total energy radiated by the Galaxy. It is therefore the most intense astrophysical radiation source at frequencies above 100 GHz.

In the case of molecular clouds, the emission is from small dust particles (with the sizes $\sim 0.1 - 10 \mu\text{m}$). Assuming a constant dust temperature T_{dust} and a RJ limit, the radiative transfer in brightness temperature is

$$T_b = T_{b0}e^{-\tau_{\text{dust}}} + T_{\text{dust}}(1 - e^{-\tau_{\text{dust}}}), \quad h\nu \ll kT$$

where $T_{b0} = 2.7 \text{ K}$ from the microwave background radiation. Usually $T_{\text{dust}} \gg T_{b0}$ and thus

$$T_b \approx T_{\text{dust}}(1 - e^{-\tau_{\text{dust}}})$$

Thermal Rad. of Interstellar Dust

For $\lambda > 100 \mu\text{m}$ (or 0.1 mm in radio regime), the dust opacity is (Mezger et al. 1990)

$$\tau_{\text{dust}} = 7 \times 10^{-21} \frac{Z}{Z_{\odot}} b N_{\text{H}} \lambda^{-\beta} \propto \nu^{\beta} \quad \text{with } \beta = 2$$

where λ is in μm , N_{H} is in cm^{-2} , Z is the metallicity as a ratio of that of the sun $Z_{\odot}$. $b \sim 1.9$ for moderate density gas and 3.4 for dense gas. In the clouds, $\tau_{\text{dust}} \ll 1$, and the brightness temperature is

$$T_b \approx T_{\text{dust}} \tau_{\text{dust}} \propto \nu^{\beta}$$

and the flux density (S or F_{ν}) is

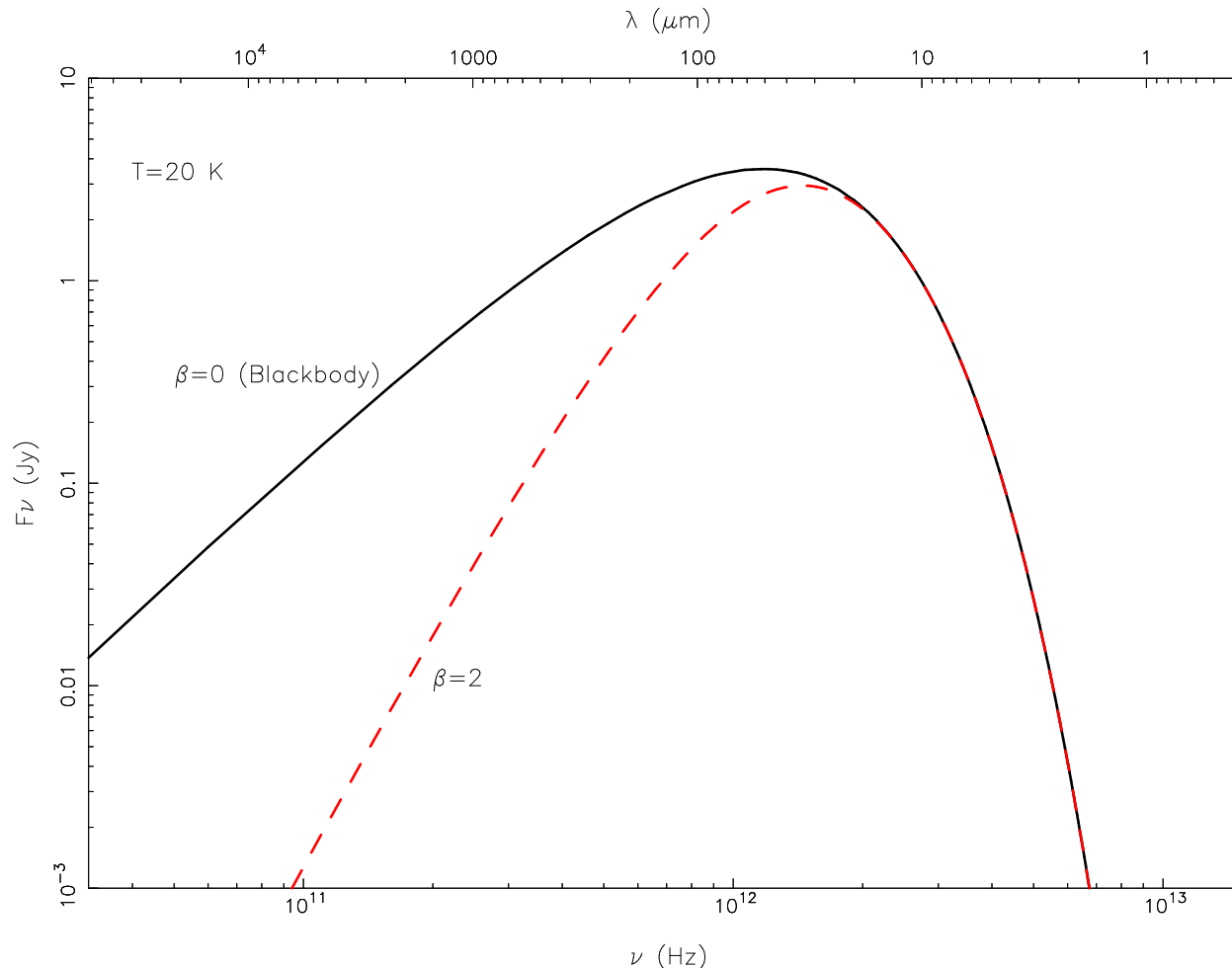
$$S = \frac{2kT_b}{\lambda^2} \Delta\Omega = \frac{2kT_{\text{dust}} \tau_{\text{dust}}}{\lambda^2} \Delta\Omega \propto \lambda^{-(2+\beta)} \propto \nu^{2+\beta}$$

Measuring S , we can determine the column density of N_{H} . If S in mJy, λ in mm, the source in FWHP sizes, θ in arc seconds, then

$$N_{\text{H}} = 1.93 \times 10^{24} \frac{S}{\theta^2} \frac{\lambda^{2+\beta}}{Z/Z_{\odot} b T_{\text{dust}}} \quad (\text{cm}^{-2})$$

It appears that cold dust (10-30 K) makes up most of the mass of dust and by implication traces cold interstellar gas (assuming a gas to dust ratio of about 100) in our galaxy.

Thermal Rad. of Interstellar Dust



The dust emission can be considered as a **modified blackbody radiation** due to the frequency (or wavelength) dependence of dust grain emissivity and thus opacity. At low frequency, the dust emission is optically thin and is only a small portion of the blackbody radiation. The dust emission is optically thick at high frequency and becomes blackbody radiation.

HII region: Ionized Gas Cloud



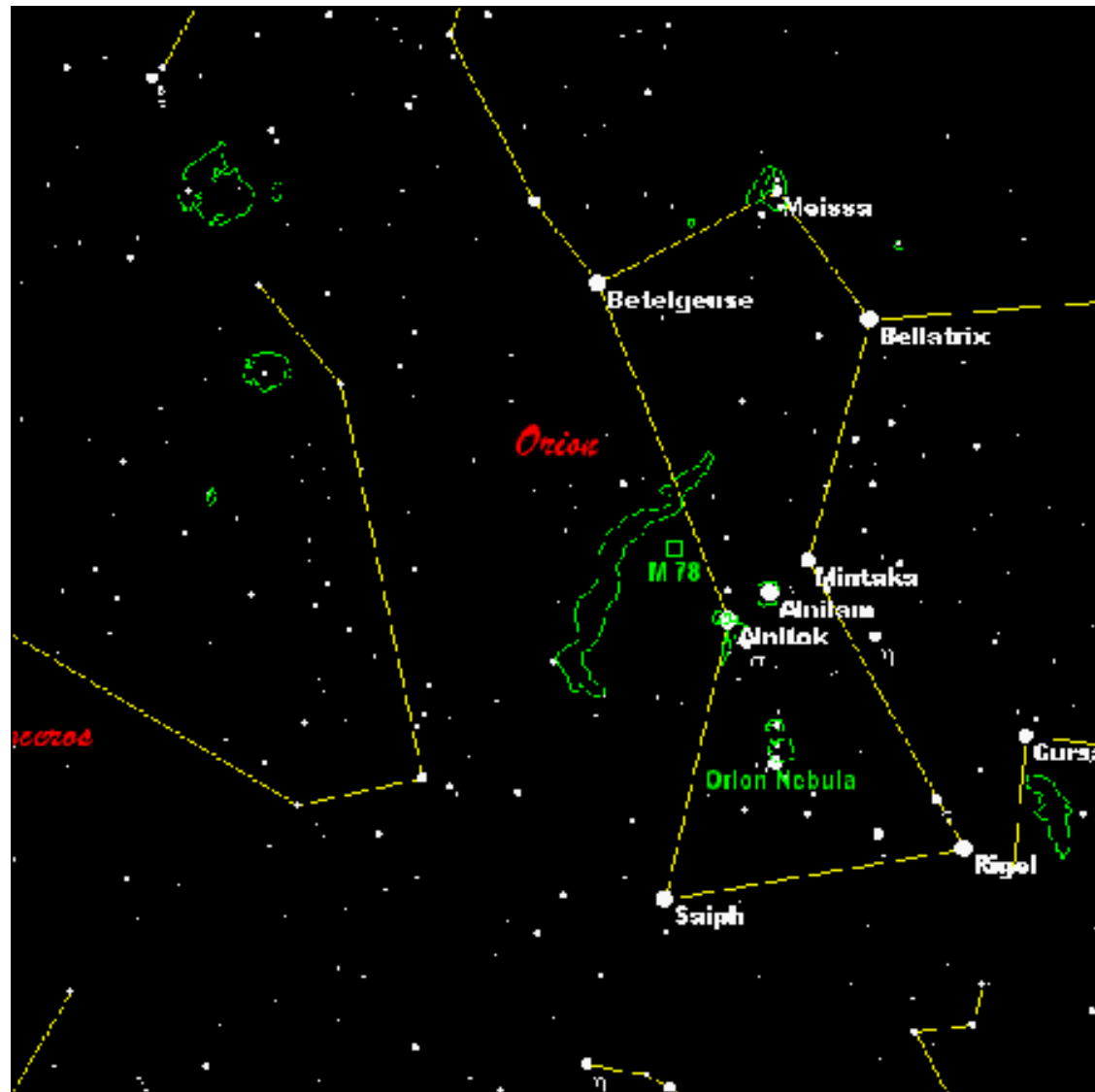
HST Image: Orion Nebula (a.k.a M42).

An H II region is a cloud of glowing gas and plasma, sometimes several hundred light-years across, in which star formation is taking place. Young, hot, blue stars which have formed from the gas emit copious amounts of ultraviolet light, ionizing the nebula surrounding them. H II regions are named for the large amount of ionized atomic hydrogen (H^+) they contain.

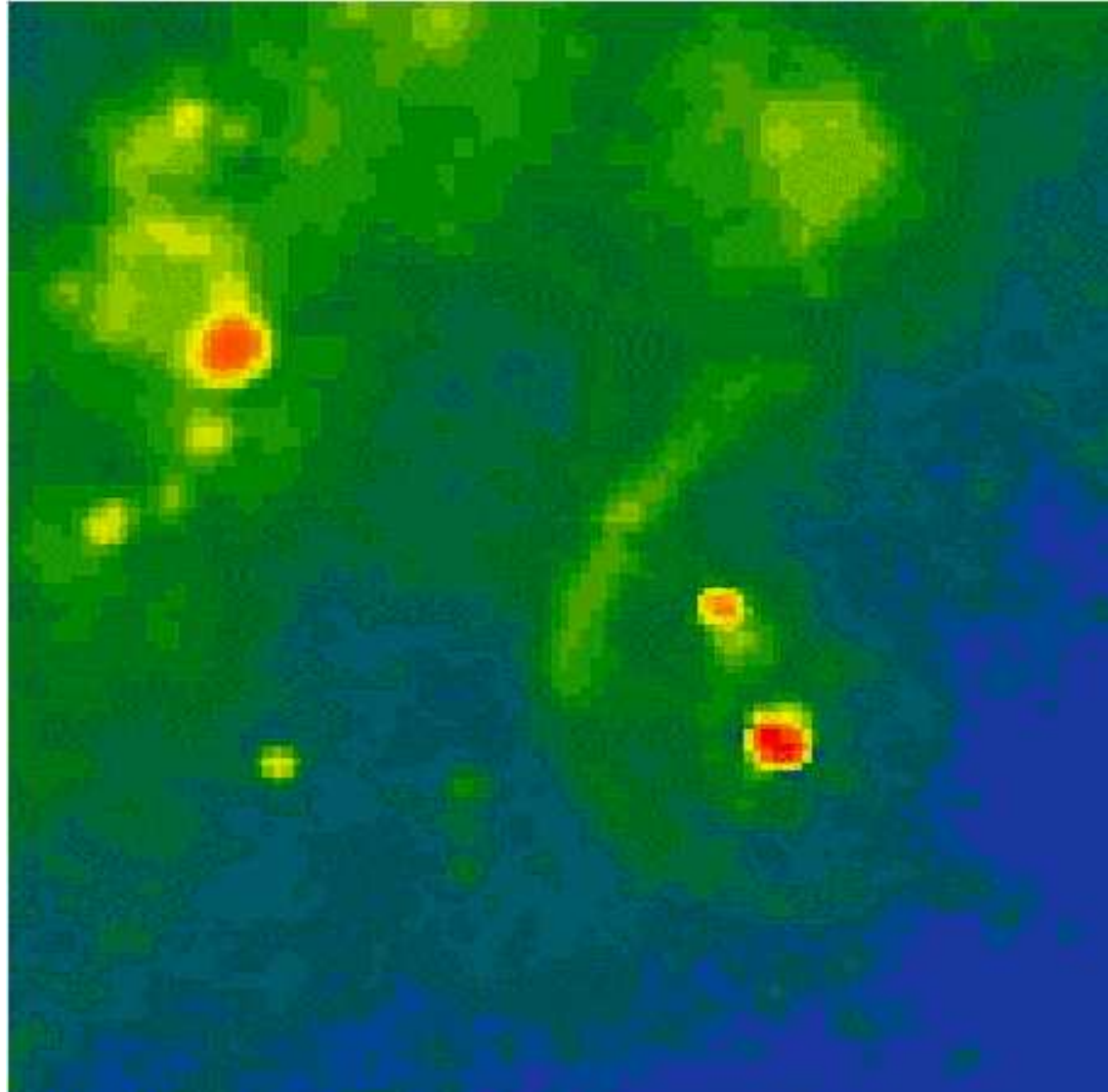
H II regions may give birth to thousands of stars over a period of several million years. In the end, supernova explosions and strong stellar winds from the most massive stars in the resulting star cluster will disperse the gases of the H II region, leaving behind a cluster such as the Pleiades.

The radio continuum in the HII region is from the electrons accelerated in the electron-ion interaction.

HII region: Ionized Gas Cloud



HII region: Ionized Gas Cloud



HII region: Ionized Gas Cloud



Optical images (left) reveal clouds of gas and dust in [the center of the Orion Nebula](#); an infrared image (right) reveals the new stars shining within.

The full details of massive star formation within H II regions are not yet well known. Two major problems hamper research in this area. First, the distance from Earth to large H II regions is considerable, with the nearest H II region being over 1,000 light-years away; other H II regions are several times that distance away from Earth. Secondly, the formation of these stars is deeply obscured by dust, and visible light observations are impossible. Radio and infrared light can penetrate the dust, but the youngest stars may not emit much light at these wavelengths.

HII region: Ionized Gas Cloud

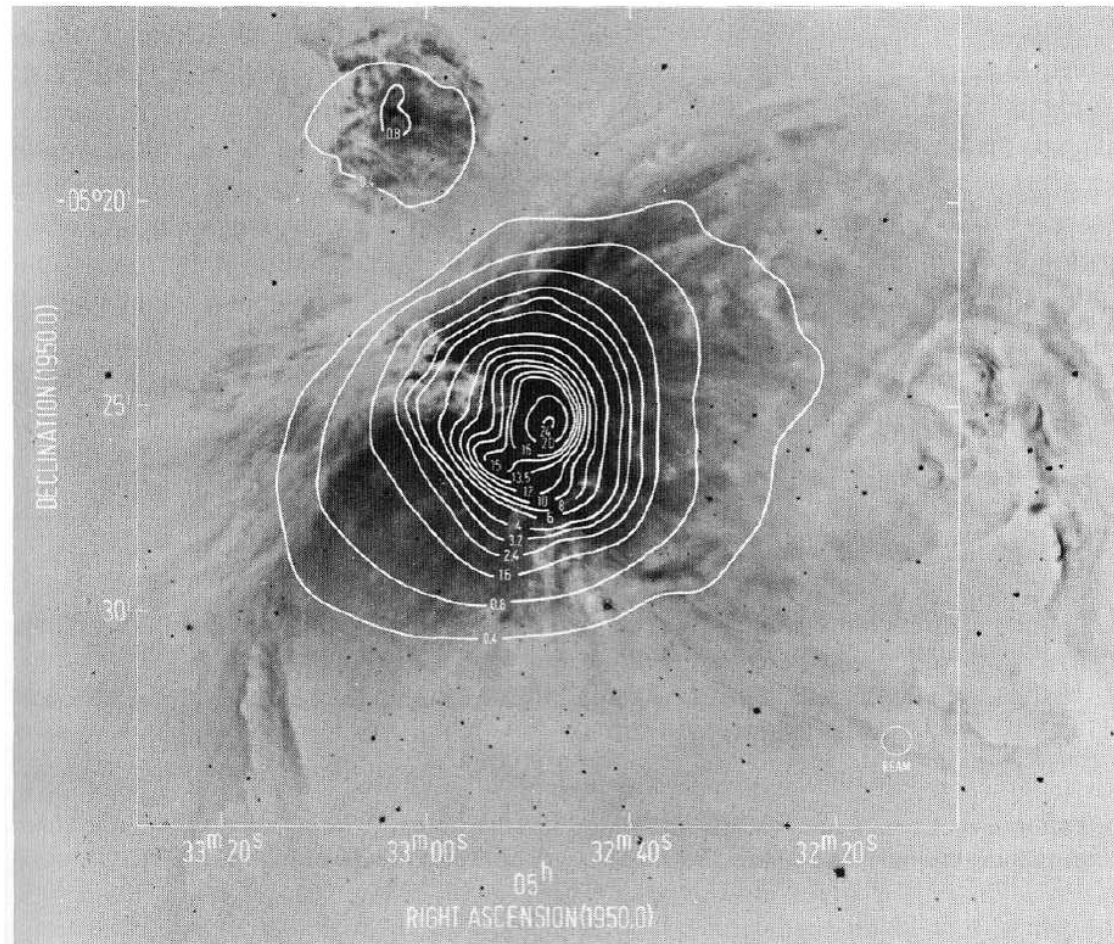


Fig. 2.3. The 23-GHz radio continuum contours, in units of main-beam brightness temperature, on an optical photo in H α and [NII] of NGC 1976 (Orion A, M42), below, and NGC 1982 (M43), above. The angular resolution is 42", which at the distance of Orion A, corresponds to a linear resolution of 0.10 pc (Wilson and Rood 1984).

HII region: SED

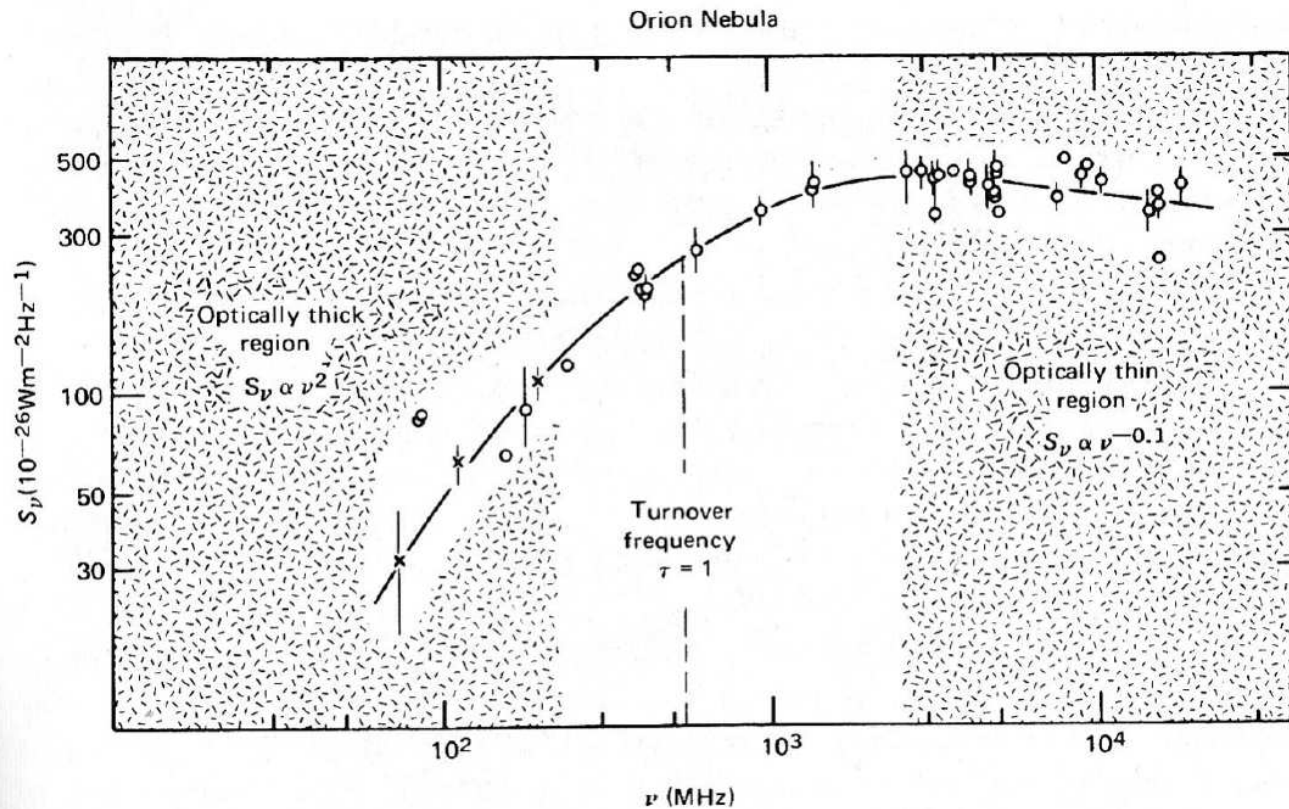
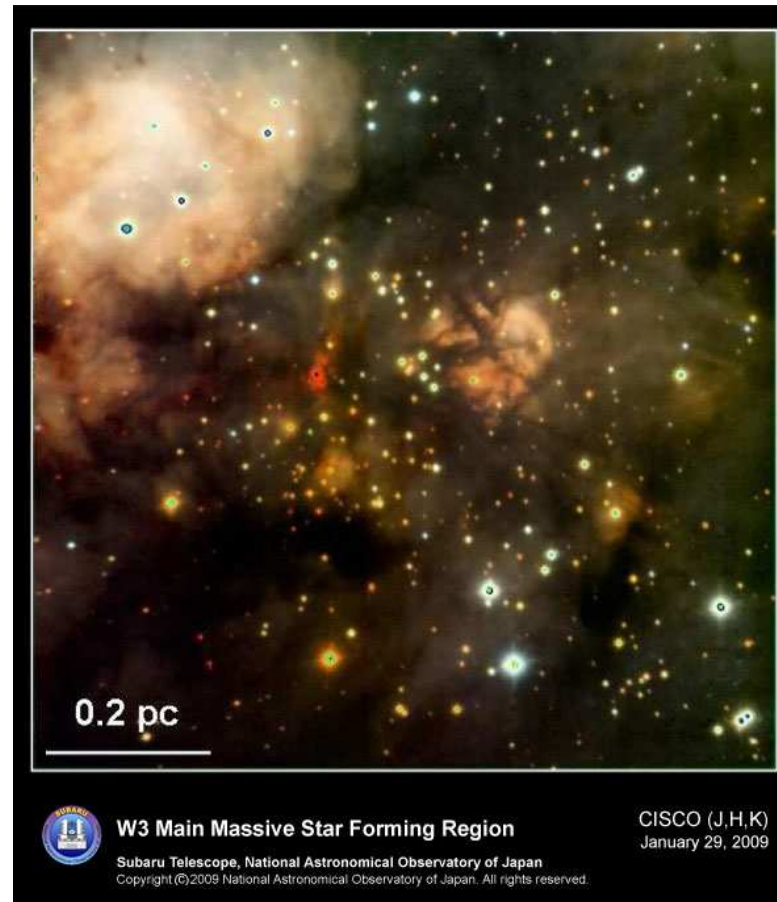


Fig. 2.2. Spectral flux density of the Orion Nebula plotted against frequency. The shaded regions mark the optically thick and thin regions of the spectrum. (Reprinted with permission by Gordon and Breach Science Publishers from: Terzian, Y. and Parrish A., *Astrophysical Letters*. Vol. 5(1970), pp. 261.

W3 Main region



W3 is a region where many massive stars are forming, located about 6,000 light years from Earth in the constellation Cassiopeia. This near-infrared image (red: K; green: H; blue: J) is the deepest and finest image from a ground-based telescope among the images of massive star forming regions. It shows distinctive reddish and bluish nebulosity features, dark filaments between the diffuse nebulosities, and a significant population of faint stars and brown dwarfs in W3 Main

HII region: Gas + Dust

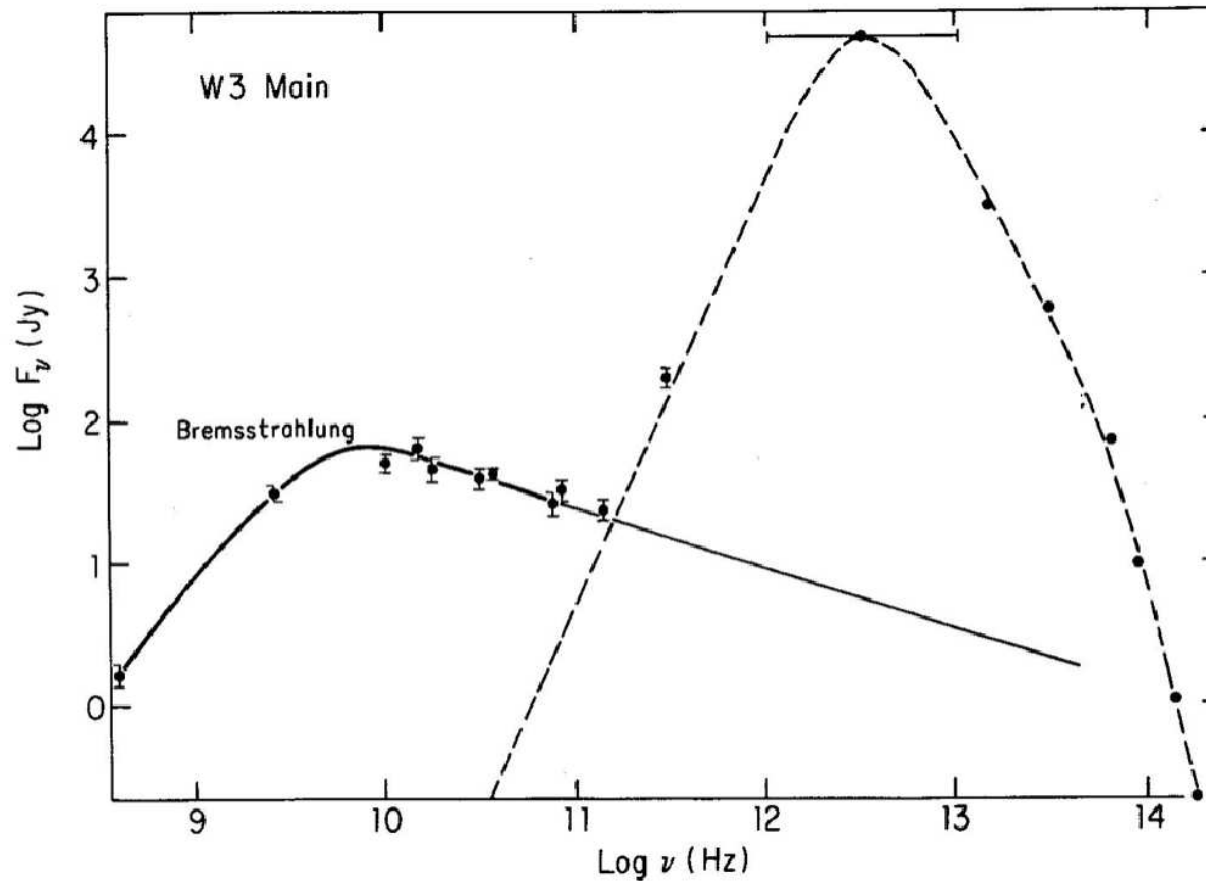
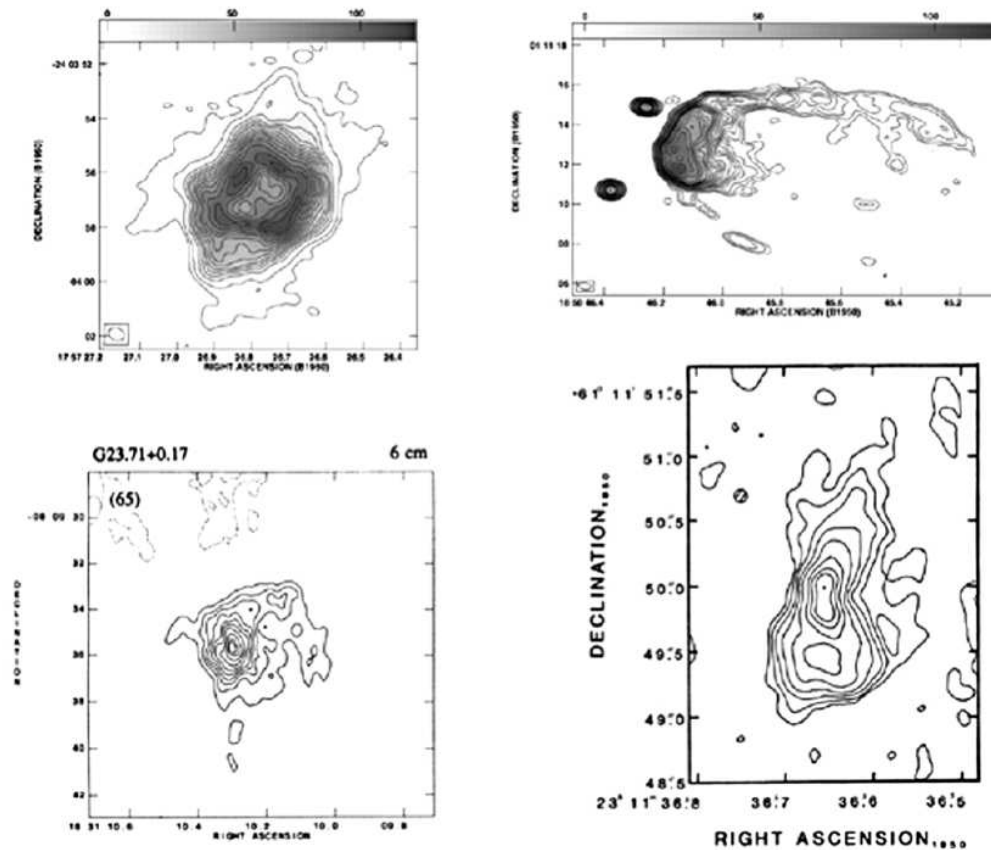


Fig. 2.4. Open circles mark observations of the integrated flux from the HII region W3 Main. Solid line: bremsstrahlung emission; dashed line: thermal emission from dust. (Malkamäki et al., 1979).

The ionized gas emission is in the radio. The dust emission is hot and peaks at the infrared, likely from the heated dust inside the HII region.

HII regions: Morphology



H II regions come in an enormous variety of sizes. Each star within an H II region ionises a roughly spherical region - known as a [Stromgren sphere](#) - of the gas surrounding it, but the combination of ionisation spheres of multiple stars within an H II region and the expansion of the heated nebula into surrounding gases with sharp density gradients results in complex shapes. Supernova explosions may also sculpt H II regions. In some cases, the formation of a large star cluster within an H II region results in the region being hollowed out from within.

HII regions in Galaxy



Strings of red H II regions delineate the arms of the Whirlpool Galaxy.

HII region: Ionized Gas Cloud

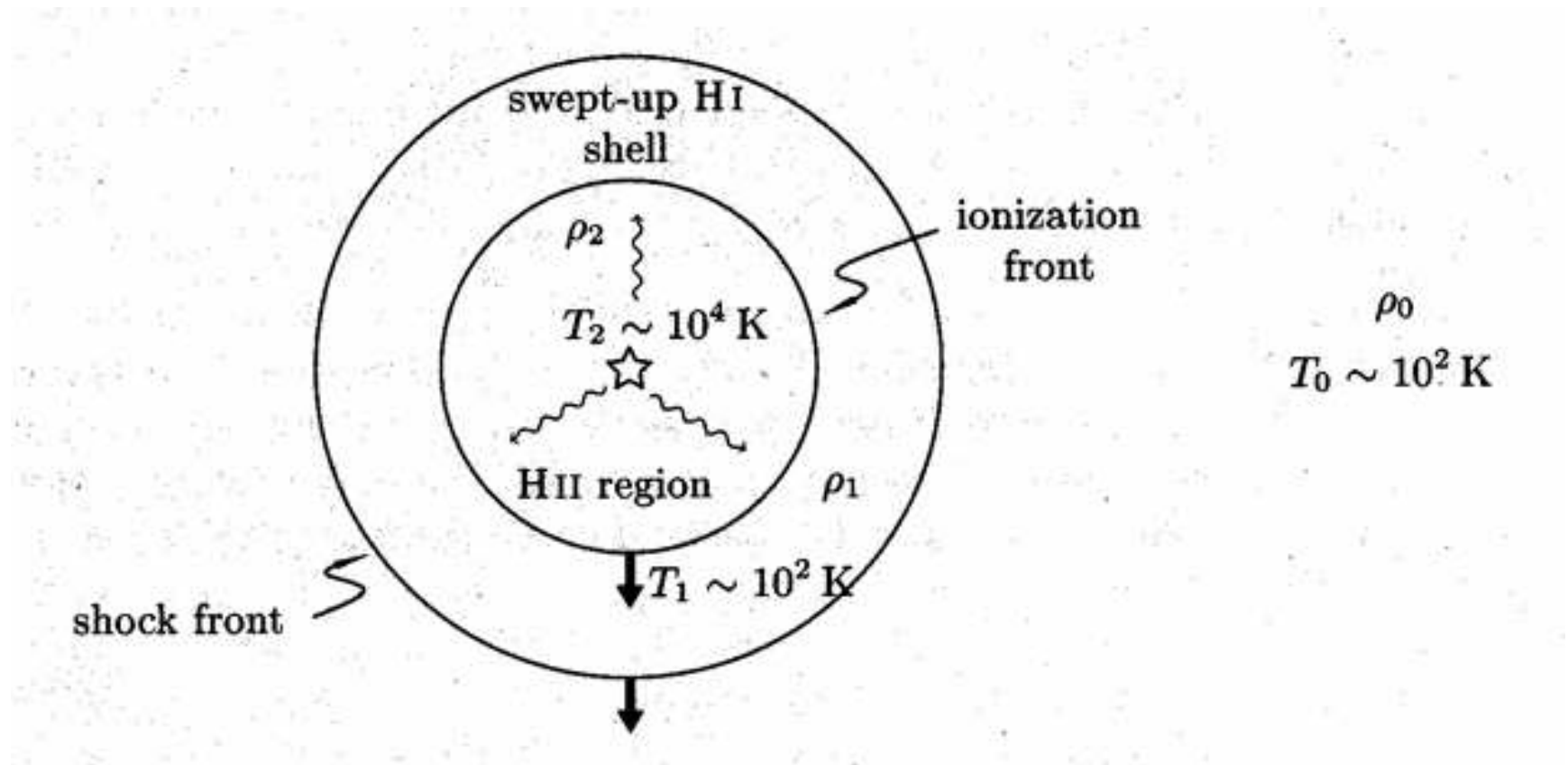
(From Wikipedia)

The precursor to an H II region is a giant molecular cloud (GMC). A GMC is a very cool (10–20 K) and dense cloud consisting mostly of molecular hydrogen. GMCs can exist in a stable state for long periods of time, but shock waves due to supernovae, collisions between clouds, and magnetic interactions can all trigger the collapse of part of the cloud. When this happens, via a process of collapse and fragmentation of the cloud, stars are born (see stellar evolution for a lengthier description).

As stars are born within a GMC, the most massive will reach temperatures hot enough to ionise the surrounding gas. Soon after the formation of an ionising radiation field, energetic photons create an ionisation front, which sweeps through the surrounding gas at supersonic speeds. At greater and greater distances from the ionising star, the ionisation front slows, while the pressure of the newly ionised gas causes the ionised volume to expand. Eventually, the ionisation front slows to subsonic speeds, and is overtaken by the shock front caused by the expansion of the nebula. The H II region has been born.

The lifetime of an H II region is of the order of a few million years. Radiation pressure from the hot young stars will eventually drive most of the gas away. In fact, the whole process tends to be very inefficient, with less than 10 per cent of the gas in the H II region forming into stars before the rest is blown away. Also contributing to the loss of gas are the supernova explosions of the most massive stars, which will occur after only 1–2 million years.

Ionization and Shock fronts



Radiation from Accelerated Electron

An accelerated Electron $\Leftrightarrow$ temporal variation in current as in Hertz dipole $\Rightarrow$ Radiation

From the Hertz dipole, we have (assuming $\mu \approx 1$ and $\epsilon \approx 1$) the radiation field

$$E_{\theta} = -i \frac{k I \Delta l}{c} \frac{\sin \theta}{r} \exp[-i(\omega t - kr)]$$

the other components are zero. Now assuming $v(t) = v_0 e^{-i\omega t}$ then $\dot{v} = -i\omega v$ and thus

$$I \Delta l = -ev = \frac{e\dot{v}}{i\omega} = \frac{e\dot{v}}{ikc}$$

Substituting this into the above eq., we have

$$E_{\theta} = -\frac{e\dot{v}}{c^2} \frac{\sin \theta}{r} \exp[-i(\omega t - kr)]$$

The electric field induced at the position of the observer by a charge at the distance r moving with the velocity $v(t)$ can be given by an expression that contains the acceleration $\dot{v}$ of the charge e at the retarded time.

Radiation from Accelerated Electron

The Poynting flux, that is, the power per surface area and steradian emitted into the direction (ϑ, φ) is

$$|S| = \frac{c}{4\pi} \mathbf{E}^2 = \frac{1}{4\pi} \frac{e^2 \dot{v}^2}{c^3} \frac{\sin^2 \theta}{r^2}$$

Integrating this over the full sphere results in the total amount of power radiated when a charge is accelerated (as for Hertz dipole):

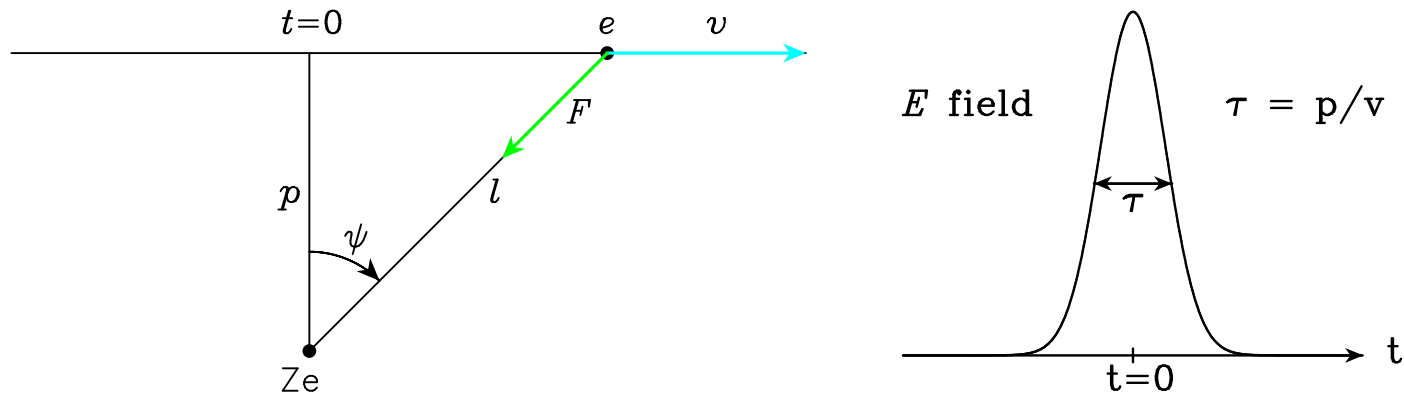
$$P(t) = \int |S| r^2 d\Omega = \frac{2}{3} \frac{e^2 \dot{v}^2(t)}{c^3}$$

The total amount of energy emitted during the whole encounter is

$$W = \int_{-\infty}^{\infty} P(t) dt = \frac{2}{3} \frac{e^2}{c^3} \int_{-\infty}^{\infty} \dot{v}^2 dt$$

Now we need to ask: [How is the electron accelerated and what is \$\dot{v}\$?](#)

Bremsstrahlung: electron + ion



Considering an electron moving past an ion of charge Ze with large distance so that the path (hyperbola in reality) can be approximated by a straight line. Let $l = p$ and $\psi = 0$ at $t = 0$ and thus

$$l = \frac{p}{\cos \psi}$$

The acceleration of the electron is given by the **Coulomb's Force**

$$F = m\dot{\mathbf{v}} = -\frac{Ze^2}{l^2} \frac{\mathbf{l}}{l} \quad \Rightarrow \quad \dot{\mathbf{v}} = -\frac{Ze^2}{ml^2} \frac{\mathbf{l}}{l}$$

For large p , F and thus $\dot{\mathbf{v}}$ and E field strength is large when ψ is small, in which the acceleration is mainly directly perpendicular to the orbit (path). In that case,

$$\dot{v} \simeq |\dot{\mathbf{v}}| \cos \psi = -\frac{Ze^2}{mp^2} \cos^3 \psi$$

Acceleration by an ion

Thus the total amount of energy emitted during the ion-electron interaction is

$$W = \frac{2}{3} \frac{e^2}{c^3} \int_{-\infty}^{\infty} \dot{v}^2 dt = \frac{4}{3} \frac{Z^2 e^6}{c^3 m^2 p^4} \int_0^{\infty} \cos^6 \psi dt$$

To evaluate the integral, we need to derive the dependence of ψ on t . [Kepler's law of areas](#): for any motion governed by a central force, the area swept per unit time by the line connecting the ion and electron is constant. Since the infinitesimal area is given by

$$dA = \frac{1}{2} l^2 d\psi$$

we have

$$\frac{dA}{dt} = \frac{1}{2} l^2 \frac{d\psi}{dt} = \text{constant} = \frac{1}{2} p v$$

with the definitions that $l = p$ and $l \frac{d\psi}{dt} = v$ at $t = 0$. Indeed $|v| \simeq \text{const}$ during the interaction. Thus,

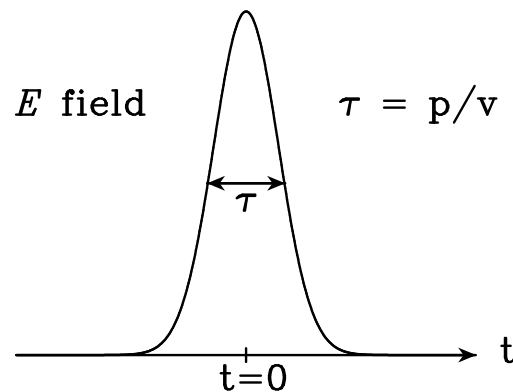
$$dt = \frac{l^2}{vp} d\psi = \frac{p}{v \cos^2 \psi} d\psi$$

Acceleration by an ion

Thus, the total amount of energy emitted is

$$W = \frac{4}{3} \frac{Z^2 e^6}{c^3 m^2 p^4} \frac{p}{v} \int_0^{\pi/2} \cos^4 \psi \, d\psi = \frac{\pi}{4} \frac{Z^2 e^6}{c^3 m^2 p^3} \frac{1}{v}$$

Now, what is the radiated frequency for this energy? **The ion-electron interaction is transient and only results in a pulse** that can be considered as a wave packet consisting of a group waves of frequency ω :



$$E(t) = \int_{-\infty}^{\infty} A(\omega) e^{-i\omega t} d\omega$$

and the wave amplitude $A(\omega)$ is given by

$$A(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} E(t) e^{i\omega t} dt$$

Here τ can be considered as the collision time. **Again Fourier Transform!**

Acceleration by an ion

The derivation of $A(\omega)$ is complicated. The result can be approximated as a step function:

$$A(\omega) = \begin{cases} 1 & \text{for } \omega \leq \omega_g \equiv \frac{3\pi^2}{16} \frac{v}{p} = 1.851 \frac{v}{p} \\ 0 & \text{otherwise} \end{cases}$$

The resulting frequency can reach up to $1.851 \frac{v}{p}$, where $\frac{v}{p} = \frac{1}{\tau}$ ($\Delta t \Delta E \gtrsim \hbar \Rightarrow \Delta \omega \gtrsim \frac{1}{\tau} \sim \frac{v}{p}$).

Thus for a given collisions parameters (p, v) , the spectrum can be approximated as follow

$$P_\nu(p, v) = \begin{cases} \frac{W}{\nu_g} = \frac{8}{3} \frac{Z^2 e^6}{c^3 m^2} \frac{1}{p^2 v^2} & \text{for } \nu \leq \nu_g \equiv \frac{\omega_g}{2\pi} = \frac{3\pi}{32} \frac{v}{p} \\ 0 & \text{otherwise} \end{cases}$$

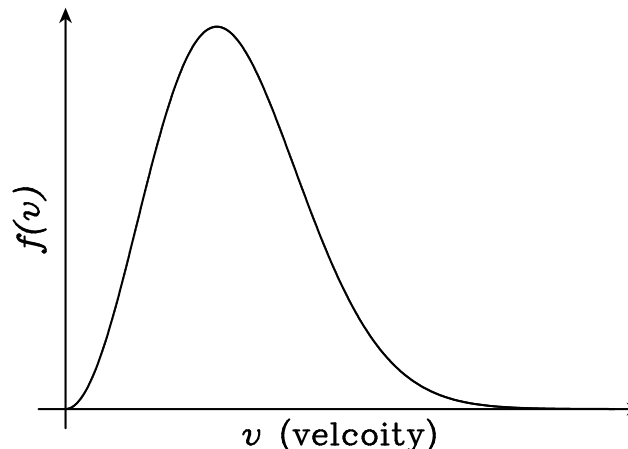
Radiation of Ionized Gas Cloud

Consider an ionized gas cloud, in which electrons have a distribution of p and v . In a thermalized medium, the **probability** dP that an electron has velocity in the velocity range $d^3\mathbf{v}$ is

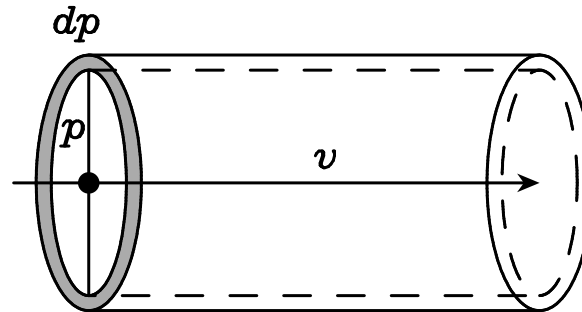
$$dP \propto e^{-E/kT} d^3\mathbf{v} = \exp\left(-\frac{mv^2}{2kT}\right) d^3\mathbf{v}$$

where m is the electron mass. Since $d^3\mathbf{v} = 4\pi v^2 dv$ for an isotropic distribution of velocities, the probability that an electron has a speed in the speed range dv is a **Maxwellian distribution**:

$$dP = \frac{4v^2}{\sqrt{\pi}} \left(\frac{m}{2kT}\right)^{3/2} \exp\left\{-\frac{mv^2}{2kT}\right\} dv \equiv f(v) dv$$



Radiation of Ionized Gas Cloud



The number of electrons in a unit volume of space with a given velocity $v = \sqrt{\mathbf{v}^2}$ that will pass by a given ion with a collision parameter between p and $p + dp$ is

$$2\pi p dp v N_e f(v) dv \quad \text{s}^{-1}$$

So number of collisions per second with collision parameters between p and $p + dp$ and velocities between v and $v + dv$ is

$$dN(p, v) = 2\pi N_i N_e v p f(v) dv dp \quad \text{cm}^{-3} \text{ s}^{-1}$$

where N_i is the number of ions per unit volume.

Radiation of Ionized Gas Cloud

As a result of these collisions, a power of $4\pi j_\nu d\nu$ will be radiated in the frequency interval ν to $\nu + d\nu$ given by

$$4\pi j_\nu d\nu = P_\nu(p, v) dN(p, v) d\nu$$

Substituting for P_ν and dN and integrating v from 0 to ∞ and p from p_1 to p_2 , then we have

$$4\pi j_\nu = \int_{p_1}^{p_2} \int_0^\infty \frac{8}{3} \frac{Z^2 e^6}{c^3 m^2} \frac{1}{p^2 v^2} 2\pi N_i N_e v p f(v) dv dp$$

where p_2 should be the mean distance between the ions or the Debye length (over which significant charge separation can occur) in the plasma. p_1 is from the quantum mechanics uncertainty principle. Thus, the emission coefficient (almost independent of ν , except for the part $\frac{p_2}{p_1}$ which has a slight dependence ==> almost flat SED at high ν !!) is

$$j_\nu = \frac{4}{3} \frac{Z^2 e^6}{c^3} N_i N_e \sqrt{\frac{2}{\pi m^3 k T}} \ln \frac{p_2}{p_1}$$

The absorption coefficient in Rayleigh-Jeans limits is

$$\alpha_\nu = \frac{j_\nu}{B_\nu(T)} = \frac{j_\nu}{\frac{2\nu^2}{c^2} k T} = \frac{4}{3} \frac{Z^2 e^6}{c \nu^2} N_i N_e \sqrt{\frac{1}{2\pi (m k T)^3}} \ln \frac{p_2}{p_1}$$

Radiation of Ionized Gas Cloud

The optical depth is

$$\tau_\nu = \int_0^s \alpha_\nu ds$$

which is integrated over the cloud and s is the size of the cloud. If the cloud is an HII region, $Z = 1$, $N_I \simeq N_e$. Assuming T is constant along the line of sight, then

$$\tau_\nu = 8.235 \times 10^{-2} \left(\frac{T}{\text{K}}\right)^{-1.35} \left(\frac{\nu}{\text{GHz}}\right)^{-2.1} \left(\frac{\text{EM}}{\text{pc} \cdot \text{cm}^{-6}}\right) a(\nu, T)$$

where $a(\nu, T) \simeq 1$ (a slow function of ν and T) and the **emission measure EM** is defined as

$$\frac{\text{EM}}{\text{pc} \cdot \text{cm}^{-6}} = \int_0^{s/\text{pc}} \left(\frac{N_e}{\text{cm}^{-3}}\right)^2 d\left(\frac{s}{\text{pc}}\right)$$

HII regions

HII can be assumed to be in LTE with constant temperature. Thus, the radiative transfer is:

$$I_\nu(\tau_\nu) = I_\nu(0)e^{-\tau_\nu} + B_\nu(1 - e^{-\tau_\nu})$$

Here $I_\nu(0)$ is the microwave background radiation and is negligible compared to B_ν at $T \approx 8000$ K for typical HII region. Thus, we have

$$I_\nu(\tau_\nu) = B_\nu(1 - e^{-\tau_\nu})$$

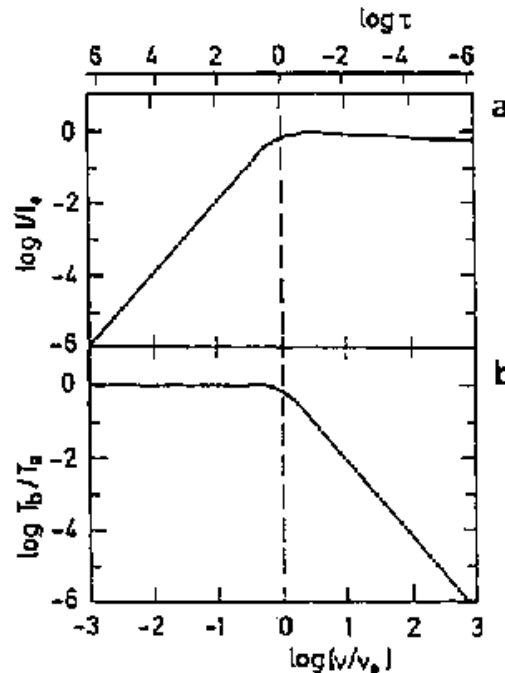
$$I_\nu(\tau_\nu) = \begin{cases} B_\nu = \frac{2\nu^2}{c^2} kT \propto \nu^2 & \text{for } \tau_\nu \gg 1 \\ B_\nu \tau_\nu (\propto j_\nu) \propto \nu^{-0.1} & \text{for } \tau_\nu \ll 1 \end{cases}$$

And $\tau_\nu = 1$ at the frequency ν_0 which is given by

$$\frac{\nu_0}{\text{GHz}} = 0.3045 \left(\frac{T}{\text{K}} \right)^{-0.643} \left(\frac{a(\nu, T) \text{EM}}{\text{pc} \cdot \text{cm}^{-6}} \right)^{0.476}$$

This frequency is called the turn over frequency at which the frequency dependency of I_ν changes from ν^2 to $\nu^{-0.1}$. For typical bright HII regions like M42, ν_0 is of the order of $\simeq 1$ GHz, EM $\approx 4 \times 10^6$ pc cm $^{-6}$ and $T \approx 8000$ K.

HII regions



Note the brightness temperature $T_b = \frac{c^2}{2\nu^2 k} I_\nu$, Thus,

$$T_b(\tau_\nu) = \begin{cases} T & \text{for } \tau_\nu \gg 1 \\ T\tau_\nu \propto \nu^{-2.1} & \text{for } \tau_\nu \ll 1 \end{cases}$$

From the observations, we measure T and ν_0 and then we can derive EM. If the size of the HII region is known, we can also derive the electron number density N_e .